Preprocessing Protocol for Electricity Theft Detection Datasets

SGCC, CER, London LCL and ETD2022

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Abstract. This protocol defines how four public electricity consumption datasets were verified, cleaned and reduced to one common daily record in kilowatt hours. It states the sources and their integrity checks, the audit performed before cleaning, the cleaning rules, the definition of a complete day in each dataset, the treatment of ETD2022, which has neither meter identity nor calendar, the quality checks, and the outputs of the preprocessing stage.

Keywords: electricity theft detection; smart meter data; data cleaning; daily load profiles; data provenance

1 Purpose and Scope

The protocol prepares consumption data for the classification of electricity theft in an offline batch setting, in which the complete series of each meter is available. It covers the work from the verified source files to the common daily table. Raw files were never modified, and every output is a derived table. Cleaning follows the supervisor's instructions. Where the research plan proposes a different treatment, the difference is recorded in Section 9.

2 Source Datasets

Table 1. Source datasets, native resolution, coverage and label scope.

Dataset	Source and version	Resolution	Meters	Observation period	Label scope
SGCC	State Grid Corporation of China research release. The local copy is identical to the GitHub release and to the Kaggle copy.	Daily kWh	42,372 consumers	1 Jan 2014 to 31 Oct 2016 (1,035 days)	Consumer level class: 3,615 theft, 38,757 not theft
CER	Commission for Energy Regulation smart metering electricity trials, data revised March 2012, six archive files.	Half hourly kWh	6,435 meters	14 Jul 2009 to 31 Dec 2010 (536 days)	None
London LCL	UK Power Networks Low Carbon London trial, London Datastore release. ACORN attributes come from the Kaggle copy.	Half hourly kWh	5,566 households, 5,561 with readings	23 Nov 2011 to 27 Feb 2014 (828 days)	None
ETD2022	Mendeley Data, version 3.	Hourly kW, ten features	16 building archetypes in 64 inferred blocks	No calendar. A nominal year is assigned	Synthetic hourly label: Normal, Theft1 to Theft6

Note. CER and ETD2022 are released under CC BY 4.0, and the London Datastore release is under a Creative Commons Attribution licence. No licence is stated for SGCC.

3 Source Verification

Table 2. Integrity checks performed on the sources.

Source	Check	Result
SGCC	SHA256 of the local copy, the GitHub release and the Kaggle copy	Identical
CER	Archive integrity (CRC) of the six files, SHA256 recorded	Passed
LCL readings	Archive integrity test of the publisher file and comparison of its ETag with the publisher value	Passed
LCL Kaggle copy	CRC of the archive and of all 114 extracted files; SHA256 of each of the 5,566 household streams against the publisher file after removing 115,453 exact duplicate rows	All matched
ETD2022	Size (64,403,651 bytes) and SHA256 against the publisher file	Matched

Note. The Kaggle copy reproduces the LCL readings exactly. The ACORN category assignments occur only in that copy and have not been verified independently against the publisher.

4 Audit Before Cleaning

Every source was audited before any rule was applied. The audit records row and meter counts, calendar coverage, cadence, missing cells and absent grid slots, zero and negative readings, duplicate keys, outlier flags and label prevalence. The results fixed the rules in Section 5 and are reported in the Summary Report.

5 Cleaning Rules

Table 3. Cleaning rules applied to all four datasets.

Rule	Treatment
Raw data	Source files are read only. All outputs are new tables, and no value in a source file is edited.
Units	Each value is energy in kWh over its native interval. A power column in kW is converted only under a stated assumption. ETD2022 hourly kW is taken as the mean power over the hour, so that kW multiplied by one hour is kWh.
Duplicates	Rows that repeat a meter, time and value exactly are removed and counted. Rows that share a meter and time but differ in value invalidate that slot and quarantine the affected day. ETD2022 has no key that defines a duplicate.
Invalid slots	Readings placed outside the expected interval grid are logged and not used. This covers off grid null rows and CER codes that cannot occur on the local day. ETD2022 rows in which every feature is zero and the archetype is 0 are quarantined.
Negative and zero values	A negative reading is invalid and treated as missing. A zero reading is valid and retained.
Short gaps	An interior run of at most three missing native intervals, with valid readings on both sides, is filled by linear interpolation and counted in `imputed_intervals`. The date 18 September 2016, which the SGCC source omits for every consumer, is treated as a one day gap in the same way.
Long gaps	A run longer than three intervals, or a run that touches the start or end of a series, stays null. Any day containing one is ineligible, and the run is logged with its meter, span, length and reason. A meter with no valid reading is ineligible throughout.
Outlier flag	For a meter with at least 30 positive readings, a reading above five times that meter's 99th percentile of positive readings is flagged. It is neither capped nor replaced.
Daily totals	A daily total is the sum of all expected intervals of the day, which are 24 for ETD2022. It is null unless every interval is valid or interpolated.
Labels	Source labels are kept together with their scope and are never altered. Synthetic labels, which only ETD2022 has, are marked by `is_injected`.

6 Common Daily Record

The four datasets are reduced to one value per meter and day, expressed in kWh. Table 4 states how a day is formed in each domain. The daily `timestamp` is the calendar day of the domain itself, and for ETD2022 it is a nominal day (Section 7). It is not a synchronized UTC instant, so the domains are comparable by duration and unit but not by moment in time.

Table 4. Definition of a complete day in each domain.

Domain	Native interval	Daily total	Calendar and clock
SGCC	Daily	The source value, unchanged	Calendar date as given in the source
CER	30 minutes	Sum of the half hourly readings of one Dublin civil day, which has 46, 48 or 50 expected slots at daylight saving changes	Local civil day. Source day code 1 is 1 January 2009
London LCL	30 minutes	Sum of 48 half hourly readings after the interval end stamps are shifted back by 30 minutes	GMT calendar day
ETD2022	60 minutes, kW	Sum of 24 hourly facility power values, each multiplied by one hour	Nominal. Relative day 0 of every block is 1 January 1971

Note. CER days with 46 or 50 slots keep their observed totals and are not rescaled to 24 hours.

Table 5. Columns of the common daily table.

Column	Type	Meaning
`domain`	Text	sgcc, cer, lcl or etd2022
`meter_id`	Text	Meter, consumer or inferred block identifier of the source
`timestamp`	Date	Calendar day of the domain, nominal for ETD2022
`kwh`	Number	Daily energy in kWh, empty when the day is not eligible
`reading_label`	Text	theft_labeled or non_theft_labeled for SGCC and ETD2022, unknown for CER and LCL
`label_scope`	Text	consumer for SGCC, day for ETD2022, none for CER and LCL
`label_source`	Text	SGCC FLAG or the ETD2022 theft column, empty for CER and LCL
`is_injected`	TRUE or FALSE	TRUE only for ETD2022, whose labels are synthetic
`analysis_eligible`	TRUE or FALSE	TRUE for a complete day
`imputed_intervals`	Number	Number of native intervals interpolated within the day
`has_high_spike`	TRUE or FALSE	TRUE when a reading of the day exceeds five times the meter's 99th percentile
`calendar_basis`	Text	source for a real calendar day, nominal for ETD2022

Note. SGCC FLAG marks a consumer that belongs to the labelled theft class. It does not state that theft occurred on a particular date. An ETD2022 day is theft_labeled when at least one of its 24 hours carries an attack label.

7 ETD2022 on the Common Basis

ETD2022 holds hourly power in kW for ten features, a building archetype in the column `Class` with 16 values, and a synthetic hourly attack label. It has no meter identifier and no calendar. Fifteen rows in which every feature is zero and the archetype is 0 are quarantined, which leaves 560,640 clean rows. These form 64 inferred blocks of 8,760 consecutive hours, four for each archetype. Each block is treated as one meter, named `etd_block_00` to `etd_block_63`, and each group of 24 hours as one day.

A nominal calendar makes the blocks usable in the common record: relative day 0 of every block is 1 January 1971 and relative day 364 is 31 December 1971. The year is arbitrary and lies far from the study periods of the other datasets, and the column `calendar_basis` marks these days as nominal. Some building types show a weekly rhythm in the data, but its weekday alignment differs between types, so the weekday and month of an ETD2022 day carry no verified meaning. A block is an inferred sequence and not a verified meter. The number of hours under each label is kept in the stage 1 daily table, because daily summation can remove the signal of an attack such as Theft6, which reverses the order of the hours.

8 Quality Checks

For each dataset the expected readings must equal the readings present plus the readings missing, and the raw rows must equal the readings present plus duplicates and invalid slots. The ETD2022 daily totals must equal the sum of the hourly values. The daily table must contain no eligible day with a null value and no negative value, and the slot counts of CER days must match the expected 46, 48 or 50. A selection of eligible London LCL daily values is compared with the sum of their 48 native intervals. The source checks of Table 2 complete the procedure. The results are reported in the Summary Report.

9 Limits of the Preprocessing

Table 6. Limits that follow from the sources and the rules.

Topic	Limit
Departure from the plan	The research plan proposes zero filling of long gaps and a three sigma cap on outliers. The supervisor's instruction takes precedence and requires exclusion and logging of long gaps and a flag at five times the 99th percentile. This protocol follows the instruction.
SGCC omitted date	The source omits 18 September 2016 for every consumer. The date is interpolated as a one day gap where both neighbours are valid, which is the case for 39,710 consumers. An interpolated day is not an observation.
Daylight saving	CER days with 46 or 50 slots keep their observed totals and are not rescaled to 24 hours.
Labels	SGCC labels describe consumers and not dated events. CER and London LCL have no theft labels. ETD2022 labels are synthetic and hourly.
ETD2022 identity and calendar	Blocks are inferred sequences and not verified meters, the calendar is nominal, and energy is estimated from power under the stated assumption.
Metadata	ACORN assignments are verified only against the Kaggle copy. The customer segment of CER is given in the publisher's allocation file and has not been joined to the outputs.

10 Outputs of the Preprocessing Stage

All paths are relative to the collection folder CER4. An Excel sheet holds at most 1,048,576 rows. A table that fits one sheet is delivered in one workbook. A larger table is delivered in consecutive numbered workbooks of at most 1,000,000 rows, each with its own header, and the parts together hold every row of the table.

Each workbook holds the cleaned daily table of one dataset, with the columns of that dataset and the values of the common layout in Table 5. The column `is_imputed` of SGCC corresponds to `imputed_intervals`, and `is_high_spike` of SGCC, CER and ETD2022 corresponds to `has_high_spike`. The columns `label_source`, `is_injected` and `calendar_basis` are constant within a dataset and are added when the four tables are combined. The combined table of 50,687,809 rows repeats the four tables and is not delivered as a separate set of workbooks.

Table 7. Workbooks of the preprocessing stage.

Workbook	Content	Rows of the table	Form
`1_preprocessing/sgcc/sgcc_daily_part01_of_44.xlsx to part44_of_44`	SGCC daily readings with raw value and flags	43,855,020	Complete, 44 workbooks
`1_preprocessing/cer/cer_daily_part1_of_4.xlsx to part4_of_4`	CER daily totals with interval counts	3,301,514	Complete, 4 workbooks
`1_preprocessing/lcl/lcl_daily_part1_of_4.xlsx to part4_of_4`	London LCL daily totals with household attributes	3,507,915	Complete, 4 workbooks
`1_preprocessing/etd2022/etd_daily.xlsx`	ETD2022 daily kWh on the nominal calendar, with hours under each label	23,360	Complete

References

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